Primer: Adversarial Inverse Reinforcement Learning (AIRL)

Fu, Luo, and Levine (2018)

econirl package documentation

1 Context and Problem

Maximum entropy IRL [Ziebart, 2010] recovers a reward function that rationalizes observed behavior, but the recovered reward entangles the true reward with the value function. For any bounded $h : \mathcal{S} \to \mathbb{R}$, the shaped reward

$$r'(s, a) \;=\; r(s, a) \;+\; \gamma\, \mathbb{E}_{s' \sim P(\cdot|s,a)}[h(s')] \;-\; h(s) \tag{1}$$

produces the same optimal policy under the original transitions P [Ng et al., 1999]. If P changes (new environment, counterfactual), the shaping term evaluates differently and the shaped reward no longer rationalizes the same behavior. Fu et al. [2018] propose AIRL to disentangle the transferable reward from the non-transferable value function by structuring the discriminator so that one component converges to $r^*(s)$ and the other absorbs $V^*(s)$.

2 Model and Notation

The agent maximizes entropy-regularized cumulative reward [Ziebart, 2010]:

$$\max_{\pi}\; \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t \big(r(s_t, a_t) + \mathcal{H}(\pi(\cdot \mid s_t))\big)\right]. \tag{2}$$

The optimal policy satisfies $\pi^*(a \mid s) = \exp(A^*)/\sum_{a'} \exp(A^*_{a'})$ where $A^*(s, a) = Q^*(s, a) - V^*(s)$, giving $\log \pi^*(a \mid s) = A^*(s, a)$.

Symbol	Meaning
$s \in \mathcal{S}$, $a \in \mathcal{A}$, s'	State, action, next state
γ, $P(s'\|s, a)$	Discount factor, transition kernel
$r^*(s)$	True reward (state-only; required for Theorem 1)
$\pi^*(a\|s)$, V^*, Q^*, A^*	Expert policy, value, action-value, advantage
$g_\theta(s)$	Reward approximator (state-only, param. θ)
$h_\phi(s)$	Shaping potential $\approx V(s)$ (param. ϕ)
$\pi_\omega(a\|s)$	Learned policy (param. ω)
$f_{\theta,\phi}(s, a, s')$	Discriminator logit (3); $D_{\theta,\phi} \in [0, 1]$: discriminator output (4)

AIRL decomposes the discriminator into reward and shaping:

$$f_{\theta,\phi}(s, a, s') = g_\theta(s) + \gamma\, h_\phi(s') - h_\phi(s), \tag{3}$$

$$D_{\theta,\phi}(s, a, s') = \frac{\exp(f_{\theta,\phi})}{\exp(f_{\theta,\phi}) + \pi_\omega(a \mid s)}. \tag{4}$$

The logit f mirrors the Bellman relationship $Q = r + \gamma\mathbb{E}[V(s')]$, decomposed into $g_\theta \approx r$ and $h_\phi \approx V$. Note g_θ depends on s only, not on a.

3 Identification

Reward ambiguity. IRL rewards are identified only up to potential-based shaping (1) [Ng et al., 1999]. The observed policy constrains $|\mathcal{S}|(|\mathcal{A}|-1)$ values but the reward has $|\mathcal{S}| \cdot |\mathcal{A}|$ unknowns, leaving $|\mathcal{S}|$ free variables corresponding to the value function [Cao and Cohen, 2021].

Decomposability (Definition B.1). An MDP is *decomposable* if all states are linked: for every pair (s, s') there exists a policy and time t such that s' is reachable from s with positive probability. This rules out absorbing states.

Disentanglement (Theorem 1). If (a) g_θ is state-only, (b) the MDP is decomposable, and (c) the adversarial training converges to the Nash equilibrium, then $g_\theta(s) = r^*(s) + c$ for a constant c.

Proof sketch. At Nash equilibrium $D = 1/2$, so $f = \log \pi^* = A^*$. Substituting (3): $g_\theta(s) + \gamma h_\phi(s') - h_\phi(s) = r^*(s) + \gamma V^*(s') - V^*(s)$. Since g_θ is state-only, taking the expectation over s' yields $g_\theta(s) = r^*(s) + h_\phi(s) - V^*(s) + c_0$. Decomposability implies $h_\phi(s) - V^*(s)$ is constant, so $g_\theta(s) = r^*(s) + c$. □

Limitations. Theorem 1 requires state-only rewards and excludes absorbing states. DDC models have action-dependent payoffs and exit options. Lee et al. [2026] resolve both with anchor constraints (see AAIRL primer).

4 Estimation Algorithm

Algorithm 1 AIRL (Fu et al. 2018, Algorithm 1)

Require: Expert demonstrations $\mathcal{D}_E = \{\tau_1, \ldots, \tau_N\}$, discount γ
Ensure: Disentangled reward $g_\theta(s)$

```
Initialize θ_0, φ_0, ω_0
for iteration = 1, 2, ... do
    D_π ← roll out π_ω to collect M transitions                    // sample from policy
    for j = 1, ..., J do
                                                                   // train discriminator
        f_n ← g_θ(s_n) + γ h_φ(s'_n) − h_φ(s_n)                    // Eq. (3)
        D_n ← σ(f_n − log π_ω(a_n | s_n))                          // Eq. (4)
        ← −E_{D_E}[log D] − E_{D_π}[log(1 − D)]
        θ, φ ← θ, φ − α_D ∇_{θ,φ}
    end for
    r̃ ← log D − log(1 − D) = f − log π_ω                           // reward signal
    Update ω via PPO on max_{π_ω} E[Σ_t γ^t r̃]                     // policy update
    if D ≈ 1/2 on held-out data then
        break
    end if
end for
return g_θ    (disentangled, transferable reward)
```

Fu et al. [2018] use PPO [Schulman et al., 2017] with $J = 1$ discriminator step per policy step. The reward signal $\tilde{r} = f - \log \pi_\omega$ is the log-odds of the discriminator. At convergence $D = 1/2$ and $g_\theta = r^* + c$ by Theorem 1.

5 Results from the Paper

Transfer experiment (Table 1 in paper). AIRL is trained on a PointMaze where the agent navigates to a goal in a room with one wall layout. The recovered reward g_θ is then used to train a new policy in a maze with different walls. AIRL transfers successfully: the agent reaches the goal in the new maze because g_θ captures the goal location, not the path around specific walls. GAIL and MaxEnt IRL fail transfer because their recovered rewards encode the original wall layout through the entangled shaping potential. This is the paper's central empirical result.

Continuous control (Table 2). On Ant, HalfCheetah, and Hopper from OpenAI Gym, AIRL achieves cumulative reward within 5% of the expert on all three tasks. GAIL matches this performance but does not recover a reward function. MaxEntIRL underperforms on the higher-dimensional tasks (Ant) because its linear reward parameterization is too restrictive for continuous control.

State-only vs. state-action (Section 5.2). When the true reward is state-only, $g_\theta(s)$ recovers it up to a constant as Theorem 1 predicts. When the true reward depends on actions, the state-only $g_\theta(s)$ absorbs the action-dependent component into the shaping potential h_ϕ. The reward is still useful for the original environment but does not transfer cleanly because the action-dependent piece is entangled with the dynamics. This motivates the AAIRL extension by Lee et al. [2026], which handles action-dependent rewards via anchor constraints.

6 Simulation

We test on a 5×5 gridworld: 25 states, 4 actions (N/S/E/W), stochastic transitions (80% intended, 20% random), $\gamma = 0.95$. True reward: $r^*(s) = 1$ at goal (bottom-right), 0 elsewhere. Expert data: 500 trajectories of 50 steps from the optimal soft Bellman policy. Run `python airl_run.py` to regenerate Table 1.

Metric	True	AIRL
Cosine similarity	− − −	0.000
Correlation	− − −	0.000
Policy match rate	− − −	0.000
Goal reward	1.00	0.000
Mean non-goal reward	0.00	0.000

Table 1: Reward recovery on 25-state gridworld (state-only reward, decomposable MDP). Auto-generated by `airl_run.py`.

The decomposability condition holds because the 20% slip probability makes all states reachable from all others. AIRL with conservative policy iteration ($\alpha_p = 0.1$) and weight decay should recover the goal reward up to a constant. Cosine similarity measures how well the recovered reward vector

aligns with the true reward vector in direction (ignoring scale). Policy match rate measures the fraction of states where the greedy action under the recovered reward agrees with the expert's greedy action. The econirl implementation uses `AIRLEstimator` with `AIRLConfig(reward_type="tabular", policy_step_size=0.1, use_shaping=True)`.

7 Comparison

	MaxEnt	MCE-IRL	GAIL	AIRL	AAIRL
Recovers reward	Yes	Yes	No	Yes	Yes
Reward transfers	No	No	N/A	**Yes**	Yes
Action-dep. r	Yes	Yes	N/A	No*	Yes
Continuous $\mathcal{S}$	No	No	Yes	Yes	Yes
Requires P	Yes	Yes	No	No	No†

Table 2: *Theorem 1 requires state-only $g_\theta(s)$. †With PPO generator; tabular variant uses known P.

AIRL's contribution over MaxEnt/MCE-IRL is reward transferability via disentanglement. The shaped reward recovered by MaxEnt IRL performs well in the training environment but fails when dynamics change, because the shaping potential h is calibrated to the original transitions. AIRL avoids this by separating g_θ (transferable) from h_ϕ (non-transferable) within the discriminator architecture.

Over GAIL, AIRL recovers a reward function, not just a policy. GAIL trains a discriminator that classifies expert vs. generated transitions, but the discriminator's internal representation is unstructured. AIRL imposes the Bellman-consistent decomposition (3), which gives the discriminator components interpretable meaning.

The main limitation is the state-only restriction. Theorem 1 requires $g_\theta(s)$ to depend on s only, not on a. In DDC models, different actions carry different payoffs (buying costs money, waiting costs time, exiting is free), so the reward is inherently action-dependent. A state-only g_θ cannot capture these differences and absorbs them into the shaping potential, breaking the disentanglement guarantee. Lee et al. [2026] resolve this with anchor constraints in AAIRL: setting $r(s, a_{\text{exit}}) = 0$ and $V(s_{\text{abs}}) = 0$ pins down the action-dependent reward uniquely, extending AIRL to the full DDC setting.

References

H. Cao and S. N. Cohen. Identifiability in inverse reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

J. Fu, K. Luo, and S. Levine. Learning robust rewards with adversarial inverse reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2018.

P. Lee, K. Sudhir, and Y. Wang. Modeling serialized content consumption: Adversarial IRL for dynamic discrete choice. Working paper, 2026.

A. Y. Ng, D. Harada, and S. Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *International Conference on Machine Learning (ICML)*, 1999.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. In *arXiv preprint arXiv:1707.06347*, 2017.

B. D. Ziebart. Modeling purposeful adaptive behavior with the principle of maximum causal entropy. In *Ph.D. Thesis, Carnegie Mellon University*, 2010.